Experimental record: first results

2026-10-18T21:20+08:00 Partial results; experiments incomplete

This edition reports observed results and retains unfinished plans. D counts candidates, n training samples per class, s signals and d selected variables.

What are we testing?

We test how few measurements retain class information, whether they remain useful after distribution change, and whether global search improves solutions enough to justify its cost.

Methods select the same number of variables and fit the same robust score. Robust risk is the largest loss when training distributions may move within the allowed distance.

What has actually been completed?

Experiment	Planned data	Recorded	Verified	Failed records
E1 synthetic selection	600	2	2	0
Original mixture anchor	500	500	500	0
E1 public data	100	0	0	0
E2 test-distribution change	120	0	0	0
E3 exact small problems	40	40	40	0
E3 scaling and cost	200	0	0	0
Small ablations	20	0	0	0

E2 reuses 120 E1 fits; timing repeats add no independent data. Dashes mean unfilled. Timeouts, failures and zero-score fallbacks remain in their denominators.

How to read the results

Weight class errors equally. Score $f(x) \in [-1, 1]$ gives probability $p_0(x) = (1 + f(x))/2$ of accepting class zero. The primary metric is

$$\mathrm{BER}_{\mathrm{prob}} = \frac{1}{2}\left[\mathrm{mean}_{Y=0}(1 - p_0(X)) + \mathrm{mean}_{Y=1}\, p_0(X)\right].$$

Lower error is better; 50% is the no-information level of a constant prediction. We also report error from classifying by the score's sign. In synthetic data, selecting h true signals gives precision h/d and recall h/s; high recall alone does not establish economical selection.

Findings supported by the observed data

- On exact small problems, global search found the independently enumerated optimum in 40/40 cases; 38/40 met the declared certificate precision. This checks optimization, rather than held-out prediction.
- On the 100-dimensional mixture, probabilistic test error is 48.41% and threshold error is 50.00%. Class-specific threshold errors are 100.0% and 0.0%, revealing a strong class bias.

E3: Search quality and the cost of certification

Use $D = 10, s = 4, d = 3$ and five or ten training samples per class: four models with five repeats give forty datasets. Enumerate 120 size-three subsets; independent transport LPs check all 176 subsets of size at most three. Forward adds one variable at a time; swaps try replacements. Feasible upper bound U and global lower bound L leave gap $U - L$ for possible improvement.

To compare search separately, fix each class's radius at its median pairwise training ℓ_1 distance divided by $4D$ (ℓ_1 sums absolute coordinate differences), with verification tolerance 10^{-8}.

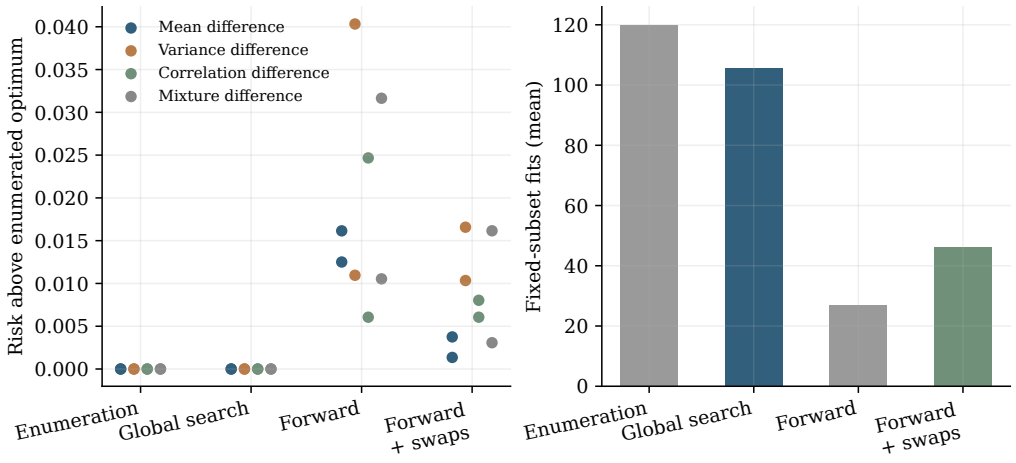


Figure 1. Left: mean excess risk per model and training-size condition (five repeats, eight points per method). Colors identify models. Independent transport LPs supply the optimum; global and enumeration overlap near zero. Right: mean fit count. Table timings reflect two concurrent jobs, not isolated hardware.

Method	Excess risk	Time / s	Fits
Enumeration	0.0000 ± 0.0000	1.06 ± 0.31	120.0 ± 0.0
Global search	0.0000 ± 0.0000	5.14 ± 1.17	105.5 ± 16.4
Forward	0.0191 ± 0.0224	0.25 ± 0.08	27.0 ± 0.0
Forward + swaps	0.0082 ± 0.0146	0.43 ± 0.19	46.0 ± 7.9

Four models and two training sizes, summarized over forty independent datasets per method. Risk is the sum of the two class losses, rather than an error percentage.

Check	Completed / planned	Failures or violations
Complete independent checks	40 / 40	0
Independent transport LPs	7,040 / 7,040	0
Lower-bound inequalities	2,101,440	0
Global interval covers optimum	40 / 40	0
Certificate target attained	38 / 40	2

Passing means no numerical counterexample was found on these instances at the stated tolerance; it does not replace a proof.

Swaps reduce mean excess risk from 0.0191 to 0.0082. Global search uses 105.5 fits versus 120 for enumeration. Concurrent timings do not establish isolated speed.

The 2 accuracy-limited cases ended at gap 1.0011e-06, just above the 0.000001 target.

Connection to the original mixture experiment

A separate anchor uses the original 100-dimensional mixtures, $X = 0.4Be + Z$ and $X = 0.4Bf + Z$. Independently draw each row's sign $B = \pm 1$ with equal probability; e is all ones, f has fifty positive and fifty negative entries, and Z is standard normal noise. Use ten training and one thousand test samples per class, five hundred repetitions and all variables. There is no unique four-variable true support.

Half the fitted robust risk bounds the worst balanced error over the specified distribution sets; test error is measured on independent observations. The threshold test assigns nonnegative scores to class zero and negative scores to class one. Classwise errors expose failures hidden by an average.

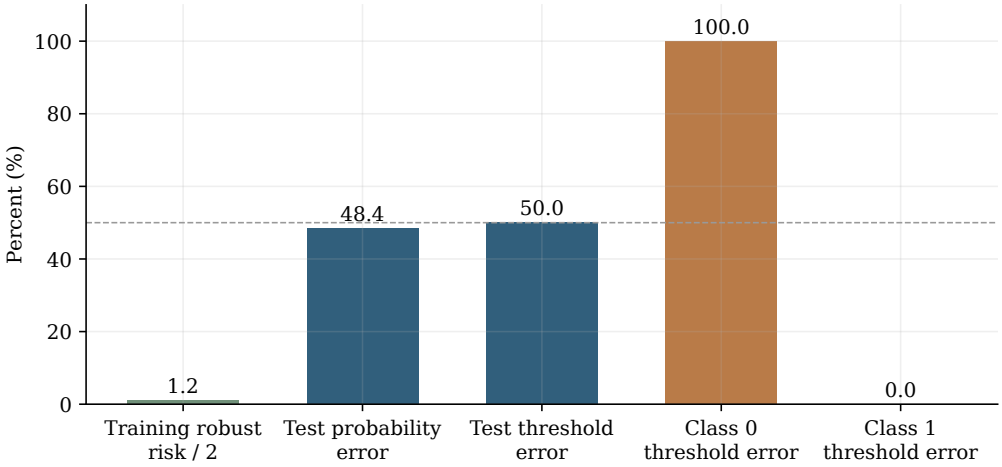


Figure 2. Original mixture: training risk and test error. Classwise errors reveal threshold predictions favoring one class; low training risk has not translated into low test error.

All-variable test	Probabilistic error	Threshold error	Observed repeats
100-dimensional mixture	48.4 ± 0.5	50.0 ± 0.0	500

The data-generating mechanism matches the original study. Our extra validation set and current score construction differ, so this is not a direct numerical replication of its table.

Across 500 repetitions, all one million test points receive the class-one threshold prediction, producing class errors of 100% and 0%. Probabilistic error is 48.41%, 1.59 percentage points below 50%, whereas half the training robust risk is only 1.16%.

The score interpolates between two training anchors by distance. A new point is typically almost as far from the class-zero anchor as the anchors are from each other. Median interpolation weights are 0.925 and 0.969 for the two true classes, favoring class one. An independent replay reproduced every saved metric. The original predictor was retained without changes after test inspection.

How data are generated and split

Each row is a record labelled $k = 0, 1$, and each column is a numerical variable. D is the candidate count, n the training count per class, s the true signal count, and d the allowed selected count. For example, $D = 50, n = 20, s = 4$ gives forty training rows and fifty columns: four signals and forty-six noise variables.

Generate each synthetic row's s signals using the rules below, then append $D - s$ independent standard normal noise variables. Draw every Z_j independently across rows and coordinates; in the mixture model, draw the row-specific sign $B = \pm 1$ with equal probability. Randomly permute columns in each repetition, sharing that permutation across splits and methods.

Mean difference	$X_j = (2k - 1)a + Z_j$.
Variance difference	Class zero: $X_j = Z_j$; class one: $X_j = rZ_j$.
Correlation difference	$X_{2h-1} = Z_{2h-1}$, $X_{2h} = (1 - 2k)\rho Z_{2h-1} + \sqrt{1 - \rho^2} Z_{2h}$.
Mixture difference	$X_{1:s} = Bav_k + Z$; v_0 is all ones, and v_1 has a positive first half and negative second half.

Before fitting methods, calibrate every model to the same difficulty: an ideal classifier knowing the true distributions and observing all signals has 20% error. In correlation and mixture data, individual columns have identical class distributions, so discrimination requires combinations. Each repetition independently draws twenty validation and ten thousand test samples per class, without copying samples across roles.

The main experiment uses $s = 4$ at three scales, $(D, n) = (20, 10), (50, 20), (100, 50)$, each with $d = 2, 4, 8$. At the medium scale also use $s = 2$ with $d = 1, 2, 4$, and $s = 8$ with $d = 4, 8, 16$. Four models and five data configurations, with thirty repetitions each, give six hundred datasets and eighteen hundred budget tasks. Budgets share the same data.

Four public datasets add real measurement conditions

Dataset	Original size	Classification task
WDBC	569 × 30	Benign versus malignant cells
Ionosphere	351 × 34	Good versus bad radar returns
Sonar	208 × 60	Rock versus metal in sonar
UCI HAR	10,299 × 561	Sitting versus standing from smartphone measurements; thirty people

For the first three datasets, split each class into a 60% training pool, 20% validation and the remainder for testing, rounding training and validation down. Draw $10, 20, 40$ training samples per class with nested sets. Use ten shared splits; compare sample sizes at budget four and budgets $1, 2, 4, 8$ at twenty training samples per class. HAR retains the official twenty-one training people and nine test people. In each of ten repetitions, reserve six training-side people for validation and sample forty windows per class evenly across the other fifteen; use budgets $8, 16, 32$. Keep alternate windows within continuous person/activity segments and weight test people equally.

Which methods are compared, and how is the comparison kept fair?

Separate the selection criterion from the search procedure. The first four procedures below optimize the same robust risk; the remaining five provide different selection criteria. Every method selects variables using training data and refits the same robust score at the same budget.

Method	What it does and why it is included
Enumeration	Fit every equal-budget subset; used only on small problems to obtain an optimal-value reference.
Global search	Our algorithm. Use risk lower bounds from evaluated subsets to reduce the search, returning a feasible solution and the remaining possible improvement.
Forward / forward with swaps	Add the variable yielding the lowest risk at each step; the latter then tries one-for-one improvements. Test whether a more local search is sufficient.
Marginal MMD	Rank columns by their class-distribution difference. MMD (maximum mean discrepancy) compares within-class and between-class similarities, detecting more than a change in the mean.
L1 logistic regression	Fit a penalized linear classifier and select the largest absolute coefficients, representing standard sparse linear selection.
mRMR	Prefer label-related variables without duplicating selected information; measure dependence by mutual information after binning at training tertiles.
Joint MMD	Compare class distributions on a group of variables, using forward steps and swaps to increase the difference; a joint criterion that can use dependencies.
Uniform random	Draw an equal-size subset with a saved seed, providing an uninformed selection reference.

Both MMD methods use Gaussian similarity with bandwidth equal to the median positive training distance on the current subset. Their score adds the two within-class mean similarities and subtracts twice the between-class mean, excluding within-class self-pairs. Joint MMD and robust forward with swaps accept at most ten improvements. All true signals and all variables are interpretive references, excluded from equal-budget rankings.

Parameter selection never accesses test data

Standardize using pooled training means and standard deviations only, dropping training-constant columns. The robust radius sets the allowed distribution movement. Its scale is the classwise median pairwise sum of absolute training differences divided by retained dimension. Compare multipliers $1/8, 1/4, 1/2, 1, 2$; at reference budget four (sixteen for HAR), select one common radius by mean validation error from marginal MMD, mRMR and joint MMD. Fix it for every method and budget. Validate L1 parameters among $0.01, 0.1, 1, 10, 100$; ties favor stronger regularization, a larger radius and a smaller column index.

Summarize synthetic results over thirty independent datasets. Compute method differences within each dataset, then use two thousand paired bootstrap resamples for 95% intervals. The ten tabular splits measure split sensitivity, rather than independent test populations. For HAR, average training repetitions within test person, then equally weight and resample the nine people; windows and repetitions do not become new people.

Remaining experiments: plan and progress

E1 retains six hundred independent synthetic datasets with three budgets each, covering four signal mechanisms, three scales and varying signal counts. Public comparisons retain the three tabular datasets and person-split HAR. E2 changes only the test distribution of E1's medium-scale four-signal, budget-four fits, using translation and measurement noise. If the matching E1 fit is unfinished, E2 is not filled by retraining a replacement.

E3 scaling centres on $D = 50, n = 20, d = 4, s = 4$, scanning $D = 20, 50, 100, 200$, $n = 10, 20, 50, 100$ or $d = 2, 4, 8, 12$, one factor at a time. Mean and mixture give two hundred independent datasets; time twenty centre instances three times and the others once. Twenty additional datasets examine half, one and twice the radius, plus one random initialization. Timing and ablation repeats do not increase independent-data counts.

Current conclusions and limits

Current evidence supports agreement between small-scale optimization and the independent reference, while exposing a substantial all-variable prediction problem. Selection gains, performance under distribution change and larger-scale efficiency each require their corresponding completed experiment.

Observed results are separated from the plan still being executed. Small-problem correctness is not statistical generalization, and the specified shifts do not cover every real change. Future score or algorithm changes require a separately identified code version and independent validation data.

Actual execution conditions

Item	Actual execution information
CPU / memory	AMD Ryzen 9 7945HX / 16 GB RAM
Algorithm / solver version	robusthw 1.0.0 / SciPy 1.18.1, HiGHS
Search time limit	Normally 300 s, HAR 600 s
Solver threads	1
Outer / inner absolute tolerances	Normally 0.001 / 0.0001; tightened for exact small problems
Numerical failures / timeouts	0 / 4

Retain formal raw records, complete condition tables and figure data together. They support verification; machine fingerprints and logs remain outside the presentation narrative.

List common radius-selection time separately and include it in total cost. Retain the best verified feasible result at timeout and zero-score fallback predictions for failures. Verified coverage on the first page refers to record integrity and the declared instance checks, not universal global-optimality certification. Development data are excluded from formal method results.